UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 007 · Taylor's and Maclaurin's expansion · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

Contents

1. Part A · Formula sheet
2. 2 · Taylor's and Maclaurin's series in one variable
3. 3 · Taylor's expansion of a function of two variables
4. 4 · Maclaurin's expansion of a function of two variables
5. Part B · Symbol glossary

A Formula sheet

2 Taylor's and Maclaurin's series in one variable

E2.1 · Taylor's series, one variable

ESA ★

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \ldots$$

GIVES	A polynomial form of f near $x = a$.
APPLIES	f has derivatives of every order at a and the series converges to f.
NOTE	Slide 007·4 writes $f^n(a)$ for the n-th derivative $f^{(n)}(a)$.

S007·4 · T007 00:04:56

E2.2 · Maclaurin's series, one variable

ESA ★

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots$$

GIVES Taylor's series about the origin.

APPLIES f and all its derivatives finite at 0 (not, e.g., $1/x$ or $\log x$).

NOTE "About a point $x = a$: Taylor; about the origin: Maclaurin."

S007·4 · T007 00:11:17

E2.3 · Exponential and logarithm

ESA ★★

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \quad (x \in \mathbb{R})$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \quad (|x| < 1)$$

GIVES Two standard series.

APPLIES As stated.

NOTE e^{-x}: replace x by $-x$.

S007·5 · T007 00:07:19 · T007 00:08:39

E2.4 · Binomial and geometric series

ESA ★

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots \qquad (|x| < 1)$$

GIVES Series for powers.

APPLIES $|x| < 1$ (any n; for a positive integer n the series stops).

NOTE The geometric series is said in the lecture, not on the slide.

S007·5 · T007 00:07:37

E2.5 · Trigonometric and hyperbolic

ESA ★★

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots, \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots, \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \ldots$$

GIVES Odd and even standard series.

APPLIES All real x.

NOTE Slide 007·5 prints $\cos x = 1 - \frac{x^2}{2} + \ldots$ ($2! = 2$).

S007·5

E2.6 · Inverse trigonometric

ESA ★

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$\sin^{-1} x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot 3}{2\cdot 4}\frac{x^5}{5} + \dots, \qquad \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

GIVES Series for the inverse functions.

APPLIES $-1 \le x \le 1$.

NOTE Slide 007·5 prints $\tan^{-1} x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$; the denominators are 3 and 5, not $3!$ and $5!$.

S007·5

3 Taylor's expansion of a function of two variables

E3.1 · Taylor's series, two variables (to second order)

ESA ★★★

$$f(x,y) = f + \frac{1}{1!}\{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots, \quad h = x-a,\ k = y-b$$

GIVES A polynomial form of f near (a,b).

APPLIES All derivatives evaluated at (a,b), all finite there.

NOTE h, k are shorthand added for clarity; the slide writes $(x-a)$, $(y-b)$.

S007·9 · T007 00:18:07 · T007 00:18:45

E3.2 · Third-order term

ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES The terms needed for "up to third degree".

APPLIES Add to E3.1 (or E4.1 with $h = x$, $k = y$).

NOTE Not on the slides; asked in four of six ESA Taylor questions.

ADDED FOR CLARITY ESA Dec-23 1b · ESA Jan-24 s1 1b · ESA Jan-24 s2 1b · ESA Jun-24 1c

E3.3 · Operator form

ESA ★★

$$f(a+h, b+k) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^n f(a,b)$$

GIVES Why the coefficients are binomial: 1, 2, 1 and 1, 3, 3, 1.

APPLIES Mixed partials equal ($f_{xy} = f_{yx}$).

NOTE Expand $(h\partial_x + k\partial_y)^n$ like $(h+k)^n$.

ADDED FOR CLARITY T007 00:18:26

4 Maclaurin's expansion of a function of two variables

E4.1 · Maclaurin's series, two variables

ESA ★★★

$$f(x,y) = f(0,0) + \frac{1}{1!}\{x f_x + y f_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}\} + \dots$$

GIVES	A polynomial form of f near the origin.
APPLIES	Derivatives at $(0,0)$, all finite there.
NOTE	Same as E3.1 with $a = b = 0$; add E3.2 with $h = x$, $k = y$ for third degree.

S007·10 · T007 00:19:38

B Symbol glossary

Symbol	Name	Meaning	Appears in
a, (a, b)	Expansion point	Point about which the series is written	E2.1, E3.1
$h,\ k$	Increments	$h = x - a$, $k = y - b$ (shorthand added for clarity)	E3.1–E3.3
$f^{(n)}(a)$	n-th derivative	Printed $f^n(a)$ on slide 007·4; not a power	E2.1
$n!$	Factorial	$1 \cdot 2 \cdots n$; divides the n-th order term	E2.1–E4.1
$f_x,\ f_y$	First partials	At the expansion point	E3.1, E4.1
$f_{xx},\ f_{xy},\ f_{yy}$	Second partials	$f_{xy} = f_{yx}$, hence the 2	E3.1, E4.1
$f_{xxx}, \ldots, f_{yyy}$	Third partials	Coefficients 1, 3, 3, 1	E3.2
n (in $(1+x)^n$)	Exponent	Any real number; not the order of a derivative	E2.4
$L(x, y)$	Taylor polynomial	Slide 007·11's name for the truncated series	W3.7

Remark. $h = x - a$ and $k = y - b$ are the shorthand used in the worked examples and are added for clarity; the slides write $(x - a)$ and $(y - b)$ in full. $f^{(n)}$ (the n-th derivative) is printed as f^n on slide 007·4; it is not a power.